

UNIT 10

Gaussian Elimination:

- Write the matrix in augmented form which includes solutions. $[A|b]$
- Perform row operations to reduce the matrix into echelon form.
- Use pivot and free variables to solve the system of linear equations.

$$Ax = b$$

$$A = LU$$

where L = lower triangular matrix

U = upper triangular matrix

Column Picture:

- This produces a parallelogram.
- Find x & y such that column vectors on LHS form the RHS, geometrically a $\square^m$.

Gauss Again:

- When breakdown occurs, it is of 2 types:
 - Permanent - cannot be solved.
 - Temporary - perform row exchange.

- If any pivot is 0 then system can either have infinite solutions or no solution
- ↓
- [imp. breakdown][permanent breakdown]

Elementary Matrices :

- Square matrix derived from identity matrix by doing one elementary row operation like
 - Row exchange
 - Row multiplication
 - Row addition.

row addition.

$E_{32} E_{31} E_{21} A = U$ row operations performed on A $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

$$A = E_{21}^{-1} E_{31}^{-1} E_{32}^{-1} U$$

where $E_{21}^{-1} E_{31}^{-1} E_{32}^{-1} = h$

$$\therefore A = LU$$

Permutation Matrix:

- type of elementary matrix
- Permutation of identity matrix [rows are swapped]
- It is used when row exchange occurs while calculating 'U' from A.

$$A = LU$$

becomes

$$\boxed{PA = LU}$$

where P = same rows swapped as was done to find U in identity form.

Transpose of Matrix:

$$\rightarrow (A_{ij})^T = A_{ji} \quad \rightarrow (AB)^T = B^T A^T$$

$$\rightarrow (A^{-1})^T = (A^T)^{-1} \text{ [if it is invertible]}$$

$$\rightarrow RR^T \text{ gives symmetric matrix}$$

$$\rightarrow RR^T \text{ may be } R^T R \text{ but not always.}$$

$$A = LU \text{ becomes}$$

$$\boxed{A = L D L^T} \text{ for symmetric matrices only. } [L^T = U]$$

or

$$\boxed{A = L D U} \text{ for any matrix}$$

Gauss-Jordan :

- To find inverse of matrix
 - Augment the matrix with its identity matrix.
 - Perform row operations to make it echelon form and then identity matrix.
 - Apply same operations to the identity matrix simultaneously.
 - When A becomes I the I will give the inverse
- $[A \mid I] \rightarrow$ perform operations on both to reduce $A = I$.

Vector Space :

- Real vector space V is set of vectors that satisfy the following conditions:
 - $\rightarrow$ if $u, v \in V$ then $u+v \in V$
[closure under addition]
 - $\rightarrow$ if $\alpha \in \mathbb{R}, u \in V$ then $\alpha u \in V$
[closure under multiplication]

Subspace :

- Subset of vector space is subspace if same operations of vector space are satisfied.

Column Space: $[C(A)]$

- contains all linear combinations of columns of A [matrix] such that $Ax = b$
- subspace of $\mathbb{R}^m$

Null Space: $[N(A)]$

- contains all vectors 'x' such that $Ax = 0$.
- subspace of $\mathbb{R}^n$

Cholesky's Decomposition:

- For square, symmetric matrix A ,

$$A = L L^T$$

where $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$ lower triangular matrix

- To calculate diagonal elements:

$$l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2}$$

- To calculate non diagonal elements $[i > j]$:

$$l_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} \cdot l_{jk}}{l_{jj}}$$